Get the slides and demos

https://github.com/AlbinGr/workshop-live-link

Agentic AI: Workflow, Risk, and Human Interaction

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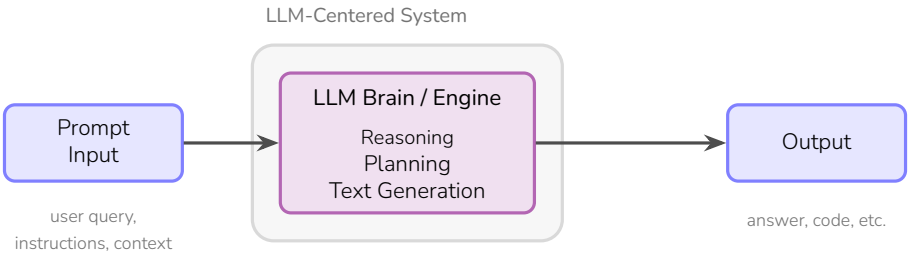
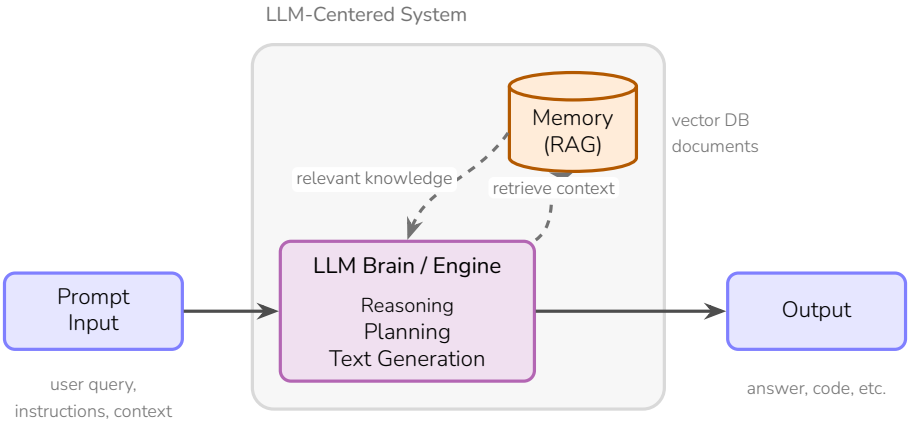

LLM-Centered System
Prompt Input
user query, instructions, context
LLM Brain / Engine
Reasoning
Planning
Text Generation
Output
answer, code, etc.


LLM-Centered System
Memory
(RAG)
vector DB
documents
relevant knowledge
retrieve context
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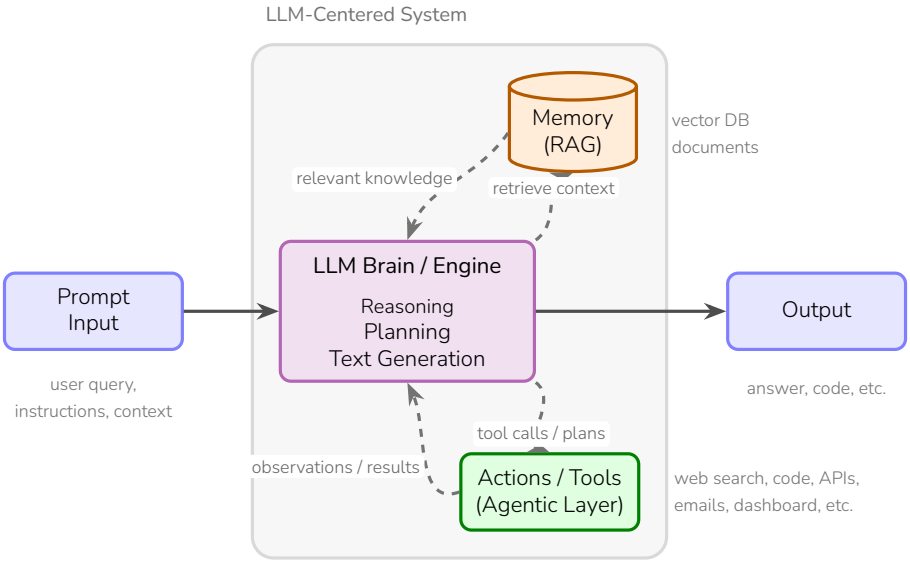
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vector DB
documents
relevant knowledge
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LLM Brain / Engine
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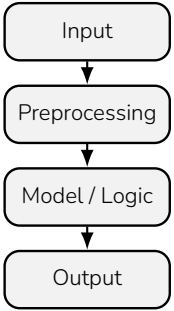
Prompt
Input
user query,
instructions, context
Output
answer, code, etc.
tool calls / plans
observations / results
Actions / Tools
(Agentic Layer)
web search, code, APIs,
emails, dashboard, etc.

Live Demo: RAG + Tool Use

- This is a minimal ReAct-style agent over a simulated email inbox.
- It combines semantic retrieval with tool helpers: search, read, flag, reply, web search, and calculator.

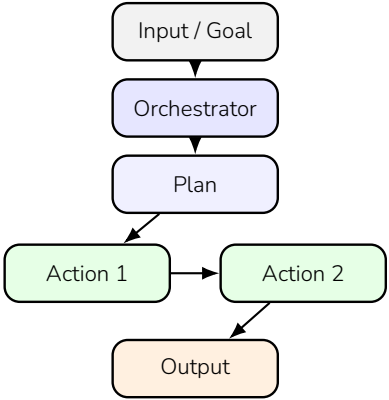
Workflow vs Orchestration

Workflow



Fixed sequence
no dynamic routing
limited adaptation

Orchestration

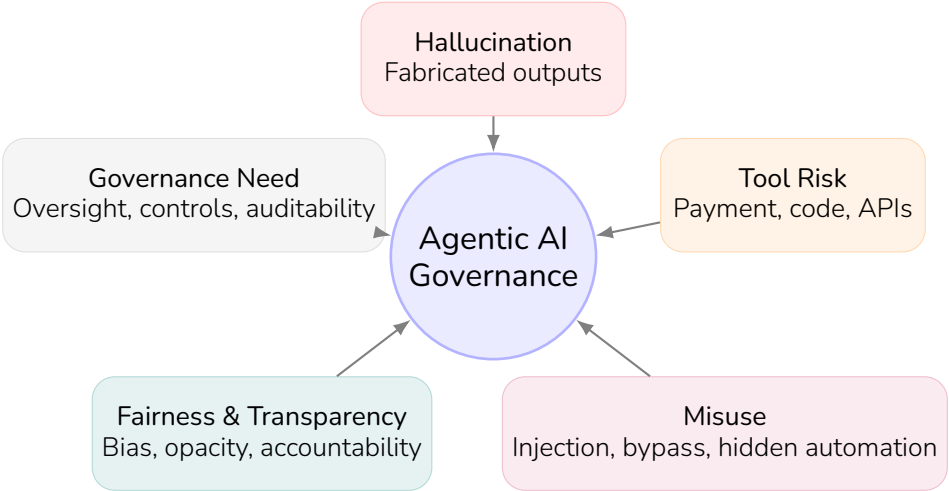


Dynamic planning followed by execution

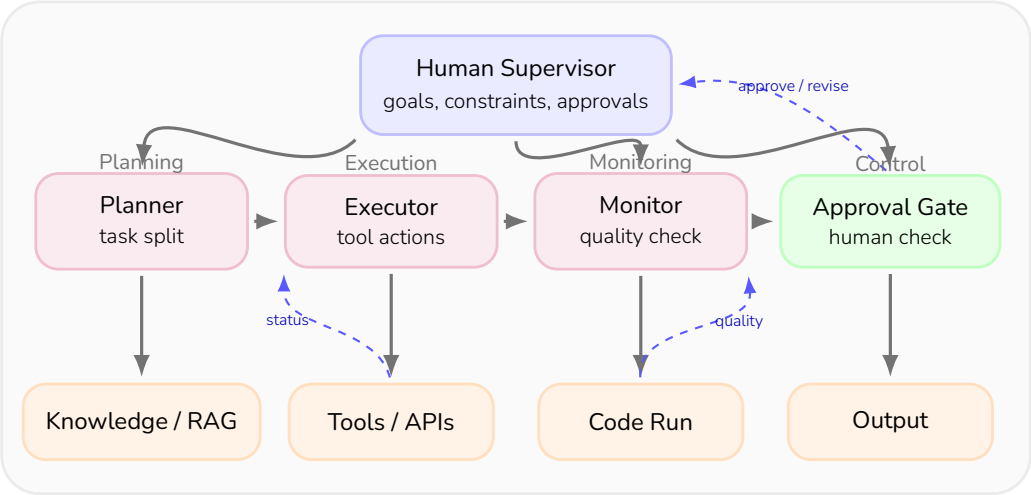
Demo: Plot generation

- A simple agent generates a plot from a user query.
- It writes Python and executes it in a safe environment with preloaded libraries such as pandas and matplotlib.
- The agent can iteratively refine the code; a workflow template variant is included for comparison.

Agentic AI Governance Risks



Human-in-the-Loop Agentic System



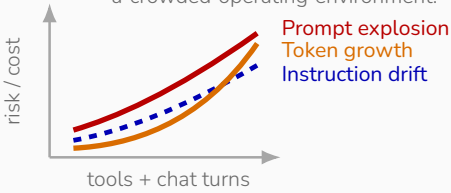
Human-in-the-Loop

- Human reviews: external communication, financial operations, compliance-sensitive actions.
- Practical intervention points:
  - before execution (approve),
  - after execution (audit),
  - confidence threshold.
- Benefit: better accountability and fewer costly mistakes.
- Trade-off: slower throughput; use selectively for critical steps.

Demo idea: ”Draft email + proposed action” -> user approves/rejects.

Agentic Scalability

As tool count and back-and-forth increase, the system prompt tends to become a crowded operating environment.



Context explosion

More tools create context and push the model away from a clean task boundary. This effect increases further with multiple chat rounds, agentic evaluation, etc.

What grows as agents grow

1. Prompt explosion with tool number

Each added tool brings descriptions, routing rules, schemas, and safety instructions. The prompt grows faster than the visible user task, so more of the model's attention is spent navigating the operating environment.

2. Instruction drift with growing context

In longer back-and-forth chats, the original operating instructions compete with follow-up edits, prior outputs, and retrieved context. That makes behavior less stable even when the base model is unchanged.

3. Large token consumption

Bigger prompts and longer histories increase latency and unit cost, while also making failures more expensive to retry. More context does not just cost more; it also increases the blast radius of mistakes.

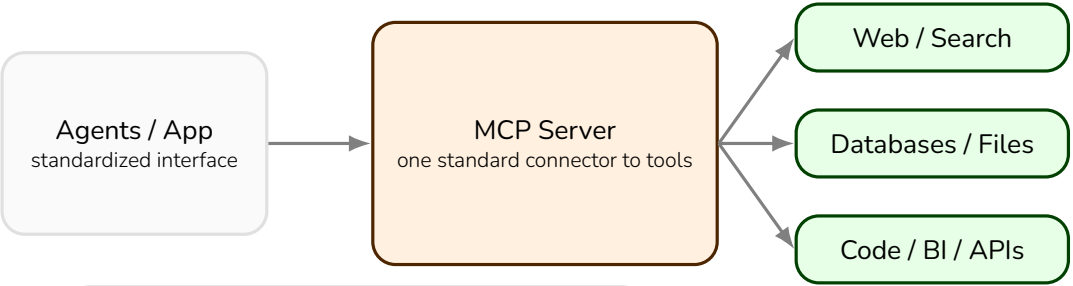
Skills: Reusable Building Blocks for Agentic AI

A skill is a reusable capability

Instead of one large fragile prompt, agentic systems work better when common tasks are packaged into small, testable units.

A skill is a short description of a specific capability that an agent can choose and then load the full context for that specific task.

MCP: The USB for Agents

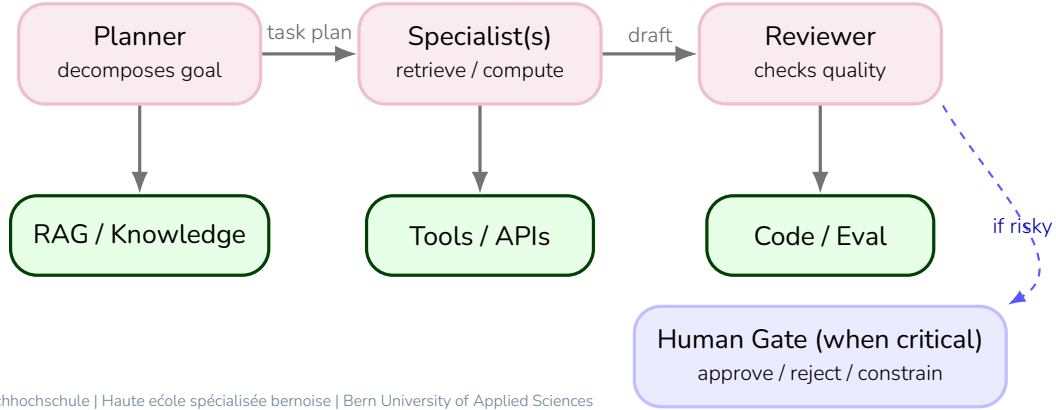


USB idea:
one protocol, many tools, fewer custom integrations

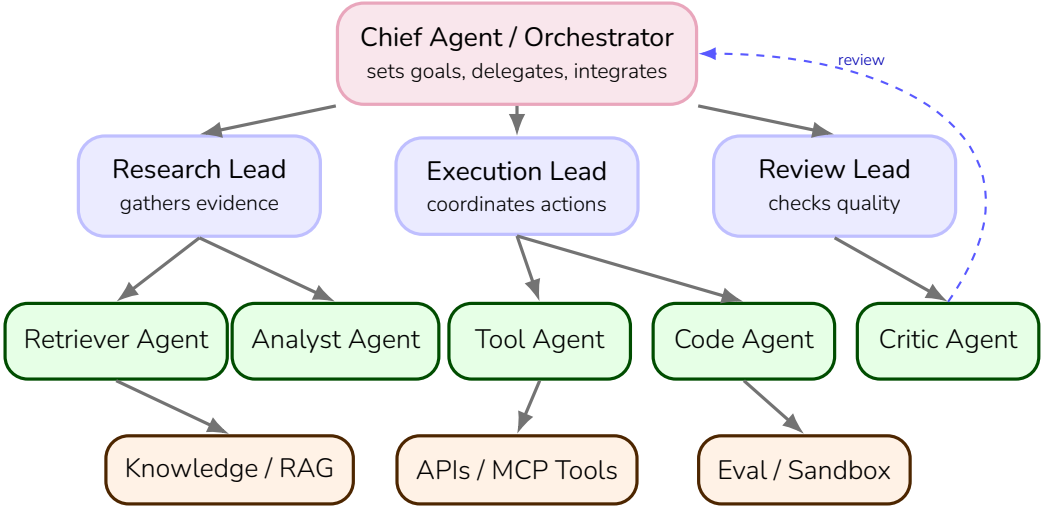
Why it matters: plug new tools in behind MCP instead of teaching every agent a new API.

Multi-Agent Pattern: Small Roles, Clear Handoffs

Key idea: Accountability comes from explicit roles + explicit interfaces.



Hierarchical Agentic System



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Machine learning in production, augmented intelligence, AI for social impact, and fairness & digital ethics.

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